

A triple-last half-word gives four-wise fair dice

Research note: exact palindrome criterion and small constructions

30 September 2026

Say that a word is triple-last fair if on every triple of letters, each die is largest with probability $1/3$.

Theorem 1. *Let V have h occurrences of each of its letters. The palindrome $W = V^{\text{rev}}V$ is three-wise fair if and only if V is triple-last fair. Whenever these conditions hold, W is also four-wise fair. No pairwise fairness assumption on V is needed.*

Proof. Fix three letters i, j, k . For an occurrence of i in V , let a, b be the numbers of earlier j 's and k 's. The corresponding two occurrences in W have prefix counts $(h - a, h - b)$ and $(h + a, h + b)$. Hence their contributions to the count of selections in which i is largest sum to

$$(h - a)(h - b) + (h + a)(h + b) = 2h^2 + 2ab.$$

Summing over all h occurrences of i gives

$$L_i(W) = 2h^3 + 2L_i(V).$$

Therefore $L_i(W) = (2h)^3/3$ is equivalent to $L_i(V) = h^3/3$. A palindrome gives the same count to an order and its reversal. For a triple, the three equal last-place probabilities, together with these reversal equalities, force all six order probabilities to equal $1/6$. This proves the equivalence.

For completeness, the standard even-degree lifting argument applies to any three-wise fair palindrome. In the completed free associative algebra, give a word its signature $S = \prod \exp(X_{\text{letter}})$. Coefficients of words with distinct letters are exactly scattered subword counts. The Baker–Campbell–Hausdorff logarithm $\log S$ is a Lie series. Reversal acts on a homogeneous Lie term of degree d by $(-1)^{d-1}$. Since the signature of a palindrome is fixed by reversal, its even-degree logarithmic terms vanish. Three-wise fairness makes its multilinear logarithmic terms of degrees two and three vanish as well. Thus the coefficient of any four distinct letters in S comes solely from the fourth power of its degree-one term, divided by $4!$. This is exactly four-wise fairness. Terms with repeated letters cannot contribute to such a coefficient. The same lifting principle is recorded in `../partial_fairness/three_four_wise_near_linear.tex`. $\square$

For the opposite orientation VV^{rev} , replace “triple-last” by “triple-first” in Theorem 1.

Corollary 1. *The minimum common face counts satisfy*

$$M_4(5) = M_4(6) = 12.$$

Proof. The exact certificates `../partial_fairness/triple_winner_half_n5_M6_seed0.json` and `../partial_fairness/triple_winner_half_n6_M6_seed0.json` give six-face triple-last fair half-words. In particular, the five-letter half-word is

001002344321324114234231132400.

The six-letter half-word is

010234432155042311324550055324114230.

For every triple and every choice of its last label, each half-word has exactly $6^3/3 = 72$ relevant selections. Their reverse-then-forward palindromes therefore give twelve-face four-wise fair dice. The lower bound follows from $M_4(n) \geq M_3(n) = 12$ for $n = 5, 6$, proved in `cloning_three_wise_dice.tex` using the six-face extension obstruction. $\square$

The independent verifier `verify_four_wise_palindromes.py` enumerates index subsets literally, without reusing the constructor's dynamic program. It checks every ordered pair, triple and quadruple: their counts are respectively 72, 288, 864. There are 120 quadruple orders in the five-letter case and 360 in the six-letter case. The verifier also checks the half-word last-place counts and the exact orientation $W = V^{\text{rev}}V$.